TARGET CAT 2025

MBA FASTRACK

QUANT : Arithmetic

Time & Work

Lecture No.- 06

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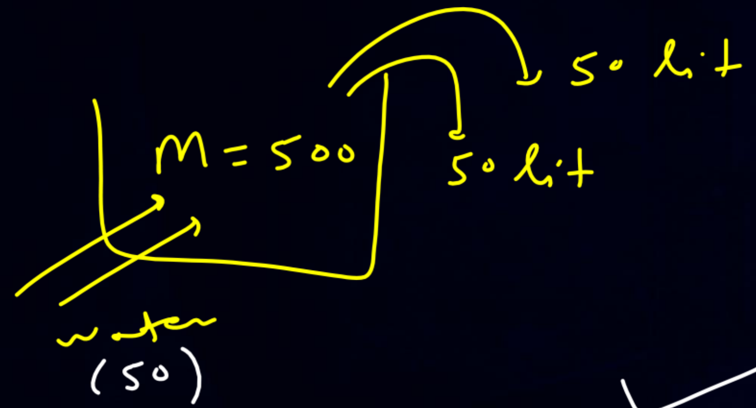
Recap of Previous Lecture :



	Removed	Remaining
I	$\frac{50}{500} = \frac{1}{10}$	$\frac{9}{10}$
II	$\frac{50}{500} = \frac{1}{10}$	$\frac{9}{10}$

$5\cancel{0}\cancel{0} \times \frac{9}{1\cancel{0}} \times \frac{9}{1\cancel{0}} = \text{New milk}$

$\boxed{405 = \text{N. milk}}$

$\text{water} = \text{Total} - \text{N. milk}$

$= 500 - 405$

$= 95$

TOPICS *to be covered*

1) Concept & Numerical based on 'Time & Work (Basics)'

2) Problems for Practice & CAT PYQs

QUESTION- 1

#Q. A glass is filled with milk. Two-thirds of its content is poured out and replaced with water. If this process of pouring out two-thirds the content and replacing with water is repeated three more times, then the final ratio of milk to water in the glass, is

[CAT 2024 : Slot 1]

let, M (original) = 81

$2/3$

Milk = 81

water

Removed → $\frac{2}{3}$

Remaining → $\frac{1}{3}$

$$\cancel{81} \times \frac{1}{\cancel{3}} \times \frac{1}{\cancel{3}} \times \frac{1}{\cancel{3}} \times \frac{1}{\cancel{3}} = \text{N. Milk}$$

$$\boxed{1 = \text{N. Milk}}$$

water = Total − N. Milk

= 81 − 1

= 80

M : W

$\boxed{1 : 80}$

(A) 1 : 27

(B) 1 : 81

(C) 1 : 26

(D) 1 : 80

Time & Work

Time & Work (Basics) :

$\boxed{\text{Effi.} \times \text{No. of Days} = \text{T.W.D}}$

$\boxed{\text{Effi.} = \frac{\text{T.W.D}}{\text{No. of Days}}}$

$\boxed{A = 15 \text{ days}}$

$\boxed{B = 20 \text{ days}}$

let, $\boxed{\text{T.W.D} = 60}$

$\boxed{\text{Effi.}}$

$A = \frac{60}{15} = 4 \text{ units/day}$

$+$

$B = \frac{60}{20} = 3 \text{ units/day}$

$A+B \rightarrow 7 \text{ units/day}$

$\text{No. of Days} = \frac{\text{T.W.D}}{\text{Effi.}}$

$= \boxed{\frac{60}{7}} \text{ days}$

Time & Work (Basics) :

Example 1:

$$\text{Effi} = \frac{T \cdot W \cdot D}{\text{Days}}$$

A, B, C and D can do a piece of work in 6, 12, 20 and 30 days respectively. Working together, they complete the work in how many days ?

$$\text{Days} = \frac{60}{20} = 3 \text{ days}$$

$$\left.\begin{array}{l} A \to 6 \\ B \to 12 \\ C \to 20 \\ D \to 30 \end{array}\right\} \text{days}$$

let, T.W.D = 60 units

Effi

$$\left.\begin{array}{l} A = \frac{60}{6} = 10 \\ + \; B = \frac{60}{12} = 5 \\ + \; C = \frac{60}{20} = 3 \\ + \; D = \frac{60}{30} = 2 \end{array}\right\} \text{Units/Day}$$

A+B+C+D = 20 units/day

Example 2:

A can lay railway track between two given stations in 16 days and B can do the same job in 12 days. With help of C, they did the job in 4 days only. Then, C alone can do the job in:

A = 16 days

B = 12 days

C = ?

A & B & C = 4 days

$\boxed{T \cdot W D = 48 \text{ units}}$

$A = \frac{48}{16} = 3$ units/day

$+ \; B = \frac{48}{12} = 4$ units/day

$+ \; \boxed{C \longrightarrow 5 \text{ units/day}}$

$A + B + C = \frac{48}{4} = 12$ units/day

$\boxed{C}$

$\text{Days} = \frac{WD}{\text{Effi.}}$

$= \frac{48}{5} = 9.6$ days

Time & Work (Basics) :

Example 3:

$$\text{Effi} = \frac{\text{T.W.D}}{\text{Time}} \qquad \text{Time} = \frac{\text{T.W.D}}{\text{Effi}} = \frac{60}{17}$$

Pipes A and B can fill a tank in 5 and 6 hours respectively. Pipe C can empty it in 12 hours. If all the three pipes are opened together, then the tank will be filled in:

let, $\boxed{\text{T.W.D} = 60 \text{ lit}}$

$+ \quad A = \frac{60}{5} = 12 \text{ lit/hr}$

$- \quad B = \frac{60}{6} = 10 \text{ lit/hr}$

$C = \frac{60}{12} = 5 \text{ lit/h}$

$A + B - C = 17 \text{ lit/h}$

Time & Work (Basics) :

$Effi = \frac{WD}{Days} \Rightarrow \boxed{Effi \times Days = WD}$

Example 4:

$\boxed{T.\ time = 'd'\ days}$

$X \to 'd'\ days$

$Y \to (d-4)\ days$

X and Y can do a piece of work in 20 days and 12 days respectively. X started the work alone and then after 4 days, Y joined him till the completion of the work. How long did the work last?

$\boxed{T.\ WD = 60\ units}$

$X = 20\ days$

$Y = 12\ days$

$X \to \frac{60}{20} = 3$

$Y = \frac{60}{12} = 5$

units/day

$\boxed{WD_x + WD_y = T.\ WD}$

$3(d) + 5(d-4) = 60$

$3d + 5d - 20 = 60$

$8d = 80$

$\boxed{d = 10}$

$\boxed{WD = Time \times Effi.}$

Example 5: let, 'A' turned off after 'x' min

Two taps A and B separately fill a cistern in 15 and 30 min, respectively. They started to fill a cistern together but tap A is turn off after few minutes and tap B fill the rest part of cistern in 9 min. After how many minutes, was tap A turned off?

$\boxed{T.\ WD = 30\ lt}$

$$A = \frac{30}{15} = 2 \quad \Big\}\ lt/min$$
$$B = \frac{30}{30} = 1$$

$$WD_A + WD_B = T.WD$$
$$2(x) + 1(x+9) = 30$$
$$2x + x + 9 = 30$$
$$3x = 21$$
$$\boxed{x = 7}\ min$$

QUESTION- 2

#Q. A can do 1/3 of the work in 4 days and B can do 2/5 of the work in 8 days. In how many days both A and B together can do the work?

$\times \frac{3}{1}$ $\times \frac{5}{2}$

$\boxed{A}$

$A = 4 \times 3 = 12$ days

$\boxed{B}$

$B = \cancel{8}^{4} \times \frac{5}{\cancel{2}} = 20$ days

$\boxed{T \cdot W D = 60 \text{ units}}$

$A = \frac{60}{12} = 5$

$B = \frac{60}{20} = 3$

units/day

$A+B \longrightarrow 8$ units/day

Days $= \frac{60}{8} = 7.5$

A) $7\frac{3}{4}$

B) $8\frac{4}{5}$

C) $9\frac{3}{8}$

D) $7\frac{1}{2}$ ✓

QUESTION- 3

#Q. If A can complete the work while working with C in 8 days. C working alone can complete the work in 12 days while working with B it can complete the work in 10 days. Find out in how many days A and B together can complete the work?

$a\left(\frac{b}{c}\right) = \frac{(a \times c)+b}{c}$

$\boxed{A \,\&\, C} \rightarrow 8$ days

$C \rightarrow 12$ days

$\boxed{B \,\&\, C} \rightarrow 10$ days

let, $\boxed{T \cdot WD = 120 \text{ units}}$

$C = \frac{120}{12} = \boxed{10 \text{ units/day}}$

$A + C = \frac{120}{8} = 15$ units/day

$A + 10 = 15$

$\boxed{A = 5 \text{ units/day}}$

$B + C = \frac{120}{10} = 12$ units/day

$B + 10 = 12$

$\boxed{B = 2 \text{ units/day}}$

$A + B = 7$ unit/day

$\boxed{\text{Days} = \frac{120}{7}}$

$17\left(\frac{1}{7}\right) = \frac{119+1}{7} = \frac{120}{7}$

A. 15

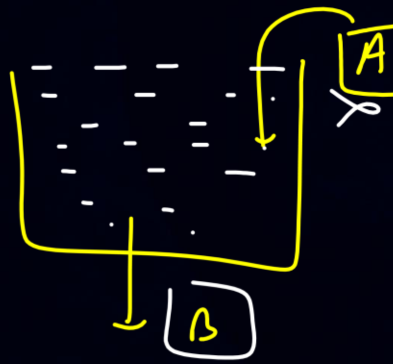
B. $\frac{7}{120}$

C. 17

D. $17\frac{1}{7}$

QUESTION- 4

#Q. A tank has an inlet pipe and an outlet pipe. If the outlet pipe is closed then the inlet pipe fills the empty tank in 8 hours. If the outlet pipe is open then the inlet pipe fills the empty tank in 10 hours. If only the outlet pipe is open then in how many hours the full tank becomes half-full? [CAT 2017 : Slot 2]



$\boxed{A = 8 \text{ hrs}}$

let, $\boxed{T \cdot WD = 40 \text{ lit}}$

(filling) $A = \frac{40}{8} = 5 \text{ lit/hr}$

$\boxed{\begin{array}{l} A = 8 \text{ hrs} \\ A \,\&\, B = 10 \text{ hrs} \end{array}}$

B (empty) $\Longrightarrow$ 1 lit/h

$A - B = \frac{40}{10} = 4 \text{ lit/hr}$

A) ~~20~~

B) 30

C) ~~40~~

D) 45

B

$\text{Time} = \frac{\text{Half WD}}{\text{eff.}} = \frac{20}{1} = 20 \text{ hrs}$

QUESTION- 5

$WD = Effi \times Days$

#Q. Anil can paint a house in 12 days while Barun can paint it in 16 days. Anil, Barun, and Chandu undertake to paint the house for ₹ 24000 and the three of them together complete the painting in 6 days. If Chandu is paid in proportion to the work done by him, then the amount in INR received by him is [CAT 2021 : Slot 3]

$\boxed{3000}$

$A = 12$ days

$B = 16$ days

$C = ?$

$\boxed{A \,\&\, B \,\&\, C = 6 \text{ days}}$

$\boxed{T \cdot Amt = Rs.\ 24000}$

Let, $\boxed{T \cdot WD = 48 \text{ units}}$

$A = \frac{48}{12} = 4$ units/day

$B = \frac{48}{16} = 3$ units/day

$C \longrightarrow \boxed{1}$ unit/day

$A + B + C = \frac{48}{6} = 8$ units/day

$A : B : C$

$(\cancel{6} \times 4) : (\cancel{6} \times 3) : (\cancel{6} \times 1)$

$\boxed{4 : 3 : 1} = 8$

$C = \frac{1}{\cancel{8}} \times \cancel{24000}^{3000} = 3000$

QUESTION- 6

T. Amt = 21000

#Q. Anil can paint a house in 60 days while Bimal can paint it in 84 days. Anil starts painting and after 10 days, Bimal and Charu join him. Together, they complete the painting in 14 more days. If they are paid a total of ₹ 21000 for the job, then the share of Charu, in INR, proportionate to the work done by him, is [CAT 2021 : Slot 2]

$A = \boxed{60}$ days
$B = 84$ days
$C = ?$

let, $\boxed{T.WD = 420}$

$A = \frac{420}{60} = 7$ units/day

$B = \frac{420}{84} = 5$ units/day

$A \rightarrow 10 + 14 = 24$ days
$B \rightarrow 14$ days
$C \rightarrow 14$ days

$\boxed{W_A + W_B + W_C = T.WD}$

$(24 \times 7) + (14 \times 5) + (14 \times C) = 420$

$168 + 70 + 14C = 420$

$14C = 420 - 238 = 182$

$C = \frac{182}{14} = 13$ units/day

$WD_A : WD_B : WD_C$

$(\cancel{24}^{12} \times 7) : (\cancel{14}^{7} \times 5) : (\cancel{14}^{7} \times 13)$

$\boxed{12 : 5 : 13} = 30$

$C = \frac{13}{30} \times \cancel{21000}^{700}$

$= 9100$

A) 9000

B) 9100 ✓

C) 9200

D) 9150

QUESTION- 7

$W = \text{Time} \times \text{Eff.}$

#Q. Two pipes A and B are attached to an empty water tank. Pipe A fills the tank while pipe B drains it. If pipe A is opened at 2 pm and pipe B is opened at 3 pm, then the tank becomes full at 10 pm. Instead, if pipe A is opened at 2 pm and pipe B is opened at 4 pm, then the tank becomes full at 6 pm. If pipe B is not opened at all, then the time, in minutes, taken to fill the tank is [CAT 2021 : Slot 2]

let. $A \rightarrow a$ lit/hr $\rightarrow$ $\boxed{5 \text{ lit/hr}}$

$B \rightarrow b$ lit/hr $\rightarrow$ $\boxed{4 \text{ lit/hr}}$

$\boxed{W_A - W_B = T.WD}$

$(8 \times a) - (7 \times b) = T.WD$

$\boxed{W_A - W_B = T.WD}$

$(4 \times a) - (2 \times b) = T.WD$

$8a - 7b = 4a - 2b$

$4a = 5b$

$\boxed{\frac{a}{b} = \frac{5}{4}}$

$(8 \times 5) - (7 \times 4) = T.WD$

$\boxed{12 = T.WD}$

$\boxed{A}$

$\text{Time} = \frac{12}{5} \text{ hrs} = \frac{12}{5} \times 60 = 144 \text{ min}$

(A) 140

(B) 120

(C) 144 ✓

(D) 264

QUESTION- 8

#Q. Gautam and Suhani, working together, can finish a job in 20 days. If Gautam does only 60% of his usual work on a day, Suhani must do 150% of her usual work on that day to exactly make up for it. Then, the number of days required by the faster worker to complete the job working alone is [CAT 2023 : Slot 3]

$\boxed{36}$

G → a units/day = 5

S → b units/day = 4

(I)

$\boxed{W_G + W_S = T \cdot W_D}$

$(20a) + (20b) = T \cdot W_D$

$\boxed{\begin{array}{l} G = 0.6a \\ S = 1.5b \end{array}}$

$(20 \times 0.6a) + (20 \times 1.5b) = T \cdot W_D$

$(20 \times 5) + (20 \times 4) = T \cdot W_D$

$180 = T \cdot W_D$

$20a + 20b = 12a + 30b$

$8a = 10b$

$\frac{a}{b} = \frac{10}{8} = \frac{5}{4}$

$\boxed{G = 5 \text{ units/day}}$

$\boxed{S = 4 \text{ units/day}}$

$\boxed{G}$

$\text{Days} = \frac{180}{5}$

$= 36$

QUESTION- 9

W.D = Time × Eff.

#Q. Renu would take 15 days working 4 hours per day to complete a certain task whereas Seema would take 8 days working 5 hours per day to complete the same task. They decide to work together to complete this task. Seema agrees to work for double the number of hours per day as Renu, while Renu agrees to work for double the number of days as Seema. If Renu works 2 hours per day, then the number of days Seema will work, is 6

[CAT 2024 : Slot 1]

T.W.D = 120 units

$R = (15 \times 4) = 60$ hrs

$S = (8 \times 5) = 40$ hrs

$R = \frac{120}{60} = 2$ units/hr

$S = \frac{120}{40} = 3$ units/hr

	Days	Hrs/day	T. Hrs
R	2d	2	4d
S	d	4	4d

$W_R + W_S = T.W.D$

$4d(2) + (4d \times 3) = 120$

$8d + 12d = 120$

$20d = 120$

$d = 6$

QUESTION- 10

#Q. Anu, Vinu and Manu can complete a work alone in 15 days, 12 days and 20 days, respectively. Vinu works everyday. Anu works only on alternate days starting from the first day while Manu works only on alternate days starting from the second day. Then, the number of days needed to complete the work is [CAT 2021 : Slot 1]

A = 15
V = 12
M = 20
days

T.W = 60 units

$A = \frac{60}{15} = 4$

$V = \frac{60}{12} = 5$

$M = \frac{60}{20} = 3$

units/day

I

V → 1, 2, 3, 4 - - - - (All days)

A → 1, 3, 5, - - - - (odd days)

M → 2, 4, 6 - - — (even days)

I → V & A → 5 + 4 = 9 units/day

II → V & M → 5 + 3 = 8 units/day

17 units/day

2 days + 2 days + 2 days + 1 day = 7 days

17 + 17 + 17 + 9 = 60 unit

- **A** 8
- **B** 6
- **C** 5
- **D** 7

QUESTION- 11

$5 \rightarrow 1$ day
$1 \rightarrow n$
0.2 day

(effi.)

#Q. Bob can finish a job in 40 days, if he works alone. Alex is twice as fast as Bob and thrice as fast as Cole in the same job. Suppose Alex and Bob work together on the first day, Bob and Cole work together on the second day, Cole and Alex work together on the third day, and then, they continue the work by repeating this three-day roster, with Alex and Bob working together on the fourth day, and so on. Then, the total number of days Alex would have worked when the job gets finished, is $\boxed{11}$ [CAT 2022: Slot 3]

B = 40 days

Effi. $\boxed{A = 2B}$

$\boxed{A = 3C}$

$6 = 3C$

$\boxed{2 = C}$

$\boxed{T \cdot W = 120}$ Units

Eff. $B = \frac{120}{40} = \boxed{3 \text{ units/day}}$

$A = 2B = 2(3) = \boxed{6 \text{ units/day}}$

$C = \boxed{2 \text{ units/day}}$

A & B $\rightarrow$ I $= 3+6 = 9$ units/day
B & C $\rightarrow$ II $= 3+2 = 5$ units/day
C & A $\rightarrow$ III $= 2+6 = 8$ units/day

22 units/day

(3 days) (3) (3) (3) (3)
I II III IV V
$22 + 22 + 22 + 22 + 22 = 110$ = 15 days

$110 + 9 = 119$

A $\rightarrow 2 + 2 + 2 + 2 + 2 + 1 = 11$

QUESTION- 12

#Q. Sam can complete a job in 20 days when working alone. Mohit is twice as fast as Sam and thrice as fast as Ayna in the same job. They undertake a job with an arrangement where Sam and Mohit work together on the first day, Sam and Ayna on the second day, Mohit and Ayna on the third day, and this three-day pattern is repeated till the work gets completed. Then, the fraction of total work done by Sam is

[CAT 2024 : Slot 2]

$\boxed{S = 20 \text{ days}}$

$\boxed{T.W.D = 60 \text{ units}}$

$S = \frac{60}{20} = 3 \text{ units/day}$

$M = 2S \implies M = 2(3) = 6 \text{ units/day}$

$M = 3A$

$6 = 3A$

$\boxed{2 = A \text{ units/day}}$

I → S & M = 9 units/day

II → S & A = 5 units/day

III → M & A = 8 units/day

22 units/day

I II III

1st 2nd 3rd

$22 + 22 + 9 + 5 + 2$

$S \rightarrow 6 + 6 + 3 + 3 + 0 = 18$

$S \rightarrow \frac{18}{60} = \frac{3}{10}$

(A) 3/20

(B) 3/10 ✓

(C) 1/20

(D) 1/5

2 Mins Summary

1) Concept & Numerical based on 'Time & Work (Basics)'

2) Problems for Practice & CAT PYQs

Thank You